

ASCII MASTER FLOW – PANEL 1/12: Weather causation chain

INSOLATION -> surface and atmospheric heating

TEMPERATURE GRADIENT -> pressure gradient

WIND -> moisture transport and convergence

LIFT + SATURATION -> cloud and precipitation

MUST REMEMBER: Weather elements connect radiation balance, temperature, pressure-gradient force, Coriolis/friction, humidity, stability, condensation, clouds and precipitation; upper-air Rossby waves, subtropical jet and western-disturbance tracks anchor Indian seasonality.

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ASCII MASTER FLOW – PANEL 2/12: Heat-budget ledger

SUN -> incoming shortwave

SURFACE -> reflected + absorbed energy

EARTH -> longwave + sensible + latent heat

BALANCE -> temperature and circulation response

ASCII MASTER FLOW – PANEL 3/12: Stability profiles

NORMAL TENDENCY -> temperature falls upward

INVERSION -> temperature rises through layer

STABLE LAYER -> vertical mixing suppressed

SURFACE INVERSION -> fog and pollution trapping

ASCII MASTER FLOW – PANEL 4/12: Wind-force triangle

PRESSURE GRADIENT -> starts acceleration

CORIOLIS -> right north / left south

FRICTION -> slows and crosses isobars

ALOFT -> near-geostrophic parallel flow

ASCII MASTER FLOW – PANEL 5/12: Planetary circulation rail

EQUATORIAL LOW -> convergence and ascent
SUBTROPICAL HIGH -> descent and trades
SUBPOLAR LOW -> westerly-polar interaction
POLAR HIGH -> cold outflow; belts migrate

ASCII MASTER FLOW – PANEL 6/12: Humidity-to-cloud chain

MOISTURE CONTENT + TEMPERATURE -> relative humidity

COOLING TO SATURATION -> dew point reached

NUCLEI + LIFT -> droplets or ice crystals

GROWTH + FALL SPEED -> precipitation

ASCII MASTER FLOW – PANEL 7/12: Rainfall comparison

CONVECTIONAL -> buoyant thermal ascent

OROGRAPHIC -> forced relief ascent

FRONTAL / CYCLONIC -> convergent dynamic ascent

REAL EVENT -> mechanisms may combine

ASCII MASTER FLOW – PANEL 8/12: Troposphere section

SURFACE -> moisture, friction and heating

MID LEVELS -> clouds, troughs and stability

UPPER TROPOSPHERE -> jets and divergence

TROPICS DEEPER / POLES SHALLOWER

ASCII MASTER FLOW – PANEL 9/12: Jet-stream engine

HORIZONTAL TEMPERATURE GRADIENT

HEIGHT / PRESSURE GRADIENT ALOFT

CORIOLIS BALANCE -> strong near-zonal wind

ROSSBY WAVES -> ridges, troughs and steering

ASCII MASTER FLOW – PANEL 10/12: India seasonal jets

WINTER -> STWJ south of Himalaya sector

STWJ -> steers western disturbances east

SUMMER -> tropical easterly jet develops

RULE -> jets are parts of wider circulation

CLOSE DISTINCTION: Humidity, relative humidity and dew point differ; weather differs from climate, jet stream from surface wind, and a western disturbance is an eastward-moving extratropical system embedded in westerlies rather than a western Indian monsoon branch.

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ASCII MASTER FLOW – PANEL 11/12: Western-disturbance chain

UPPER TROUGH / LOW -> eastward westerly motion

MOISTURE + DYNAMICAL LIFT -> cloud and rain/snow

HIMALAYA + PLAINS -> terrain-specific impacts

OUTCOME -> beneficial or damaging by timing/intensity

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ASCII MASTER FLOW – PANEL 12/12: Forecast evidence ladder

RADIOSONDE / PILOT BALLOON -> upper-air state

SATELLITE + RADAR -> cloud and precipitation

NWP ENSEMBLE -> scenarios and uncertainty

WARNING -> action; later observations verify outcome